Md. Rakib Rayhan

Undergraduate Student, Computer Science and Engineering · **Research Assistant**, MESIL Lab, NSU
Computer Architecture · RTL Design & Verification · FPGA Systems
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PROFILE

Final-year Computer Science and Engineering student at North South University and **Research Assistant** at the Microprocessor, Embedded Systems, and IoT Lab (MESIL). I design processors in RTL and verify them on **FPGA hardware**, and have built **MIPS, RISC-V and ARM cores from single-cycle to pipelined** on a low-cost FPGA board. Seeking higher study in **computer architecture, RTL design and FPGA systems**.

EDUCATION

BSc in Computer Science and Engineering 2022 – 2026 (expected)
Department of Electrical and Computer Engineering, North South University, Bangladesh

- **Thesis:** RTL Design and Hardware Verification of a Multi-Cycle Processor on a Low-Cost FPGA Development Board (defended 2026)
- **Supervisor:** Dr. Mohammad Abdul Qayum, Assistant Professor, Dept. of ECE
- **Relevant coursework:** Computer Organization and Architecture, Microprocessor Interfacing & Embedded System, Operating Systems Design, Data Structures and Algorithms, Database Systems, Artificial Intelligence, Machine Learning, Pattern Recognition and Neural Network

RESEARCH EXPERIENCE

Research Assistant Sep 2026 – Present
Microprocessor, Embedded Systems, and IoT Lab (MESIL), Dept. of ECE, North South University
Supervisor: Dr. Mohammad Abdul Qayum

- Design, implement and verify processors in RTL on **FPGA hardware**: **MIPS, RISC-V (RV32I, RV32IM) and ARM** cores in single-cycle, multi-cycle and pipelined forms.
- Lead **microarchitecture comparison studies** of ARM, RISC-V and MIPS cores, all built on the same RZ-EasyFPGA A2.2 board (Intel Cyclone IV E, EP4CE6E22C8); **journal paper in preparation**.
- Fit each core within the board's very limited logic-element budget while keeping the hardware standard, correct and verifiable.

PUBLICATIONS

Md. Rakib Rayhan, Mohd Atik Akbar Rahat, Adnin Hossain, A F Dhrubo, Mohammad Abdul Qayum, Mohsin Sajjad. *Design and Hardware Verification of a 32-Bit Multi-Cycle MIPS Processor on a Low-Cost Cyclone IV FPGA*. Submitted to the International Conference on Computer and Information Technology (ICCIT 2026).

PROCESSOR DESIGN PROJECTS

Microarchitecture Comparison Studies: ARM, RISC-V and MIPS MESIL research

- Comparing the behavior of single-cycle, multi-cycle and pipelined cores across ARM, RISC-V and MIPS on one low-cost FPGA board; ongoing research at MESIL Lab, NSU.

ARM Processors (ARMv4 subset) on FPGA Verilog · Cyclone IV

- Built single-cycle, multi-cycle and pipelined ARM cores on the same board as the RISC-V study, for ARM-versus-RISC-V comparison on identical hardware.

RISC-V Processors (RV32I, RV32IM) on FPGA Verilog · Cyclone IV

- Built a single-cycle RV32I core and multi-cycle and pipelined RV32IM cores with multiply and divide; the main challenge was fitting full designs into very limited logic elements.

MIPS Processors on FPGA Verilog · Cyclone IV

- Built single-cycle, multi-cycle (undergraduate thesis) and pipelined MIPS cores; the thesis core was presented at the Bangladesh Innovation Fair 2026.

32-bit Single-Cycle MIPS Processor with Operating System Verilog · C++

- Designed a 32-bit single-cycle processor on a custom MIPS-like ISA (including JAL and JR), with a C++ assembler and a minimal operating system in MIPS assembly; verified in ModelSim.

23-bit Single-Cycle MIPS CPU Logisim · Python

- Designed a single-cycle CPU in Logisim around a self-defined 23-bit MIPS-inspired ISA, with the complete datapath, control unit and ALU control logic, and a Python assembler that produces hex machine code for the instruction memory.

HARDWARE ACCELERATOR PROJECTS

Fixed-Point Scaled Dot-Product Attention Accelerator Verilog · Icarus · Python

- Designed an RTL accelerator for transformer attention, $\text{softmax}(QK^T/\sqrt{d_k})\cdot V$: one shared 18×18 MAC driven by an FSM, Q8.8 fixed-point data, a **LUT-based base-2 softmax** with one reciprocal per row, and row streaming for O(N) score memory.
- Verified **bit-exact** against a Python fixed-point model in Icarus Verilog, within 1.5 LSB of floating point.

Iterative NTT/INTT Accelerator for Lattice-Based Cryptography Verilog · Icarus · Python

- Designed a number-theoretic transform core with a single reused butterfly, a **Barrett modular multiplier**, a twiddle ROM and an FSM address generator, parameterized from N = 8, q = 17 up to N = 256, q = 7681.
- Verified forward and inverse transforms against the direct formula, plus polynomial multiplication on hardware.

OTHER PROJECTS

- **Knowledge distillation for rice variety classification:** teacher–student framework for 20 Bangladeshi rice varieties; ResNet50 teacher **98.59%** accuracy, lightweight CNN student **96.06%** with **13.31× fewer parameters**; Grad-CAM analysis.
- **Psychological distress self-assessment:** hybrid machine-learning approach classifying anxiety, stress and depression levels in university students, comparing 15 algorithms.
- **Custom memory allocator:** thread-safe malloc/free in C on Linux system calls, with an optimized block-reuse strategy, benchmarked under multi-threaded load.
- **SneakPick:** Django e-commerce application with catalog, accounts, cart, checkout, order tracking and reviews.
- **Football Club Management System:** full-stack PHP and MySQL web application with secure authentication, modules for clubs, managers, coaches, players, matches and stadiums, and an admin dashboard.

LEADERSHIP & ACTIVITIES

Vice Chair (Technical), IEEE North South University Student Branch Apr 2025 – Apr 2026

- Organized **more than a hundred** technical events and activities for students.
- Key events organized: **IEEE Student Transition and Elevation Partnership (STEP) 2025: Igniting Future Innovators**; **Student Professional Awareness Conference (SPAC-25)**; the national **Future Builders: AI-Driven Cognitive Innovation Hackathon**; **Skill Development Venture** (a series of events); and **IEEE Day** (a series of events).
- During this tenure, the branch received the **2026 IEEE Region 10 Outstanding Student Branch Award**.

Bangladesh Innovation Fair 2026 2026

- Represented North South University as a Research Assistant at MESIL, presenting FPGA processor work.

BD Apps National Hackathon 2023 2023

- Selected for the regional round.

TECHNICAL SKILLS

HDL	Verilog, VHDL
EDA tools	Intel Quartus, ModelSim, Questa, Logisim
Architectures	MIPS, RISC-V (RV32I, RV32IM), ARM (ARMv4 subset)
FPGA	Intel Cyclone IV E (RZ-EasyFPGA A2.2), Quartus synthesis, on-board debugging (ISSP, In-System Memory Content Editor)
Programming	C, C++, Python, Java, PHP, MIPS assembly
Web	HTML, CSS, MySQL, Django
Machine learning	CNNs, knowledge distillation, classical ML pipelines
Languages	English (fluent), Bangla (native)

REFERENCES

Dr. Mohammad Abdul Qayum Thesis and Research Lab Supervisor

Assistant Professor, Department of Electrical and Computer Engineering, North South University

Ph.D. in Electrical Engineering (Computer Engineering), New Mexico State University, USA

Email: mohammad.qayum@northsouth.edu *Other references available on request.*